Bouquet: A Visualization Tool for Symmetry Sets and Vineyards

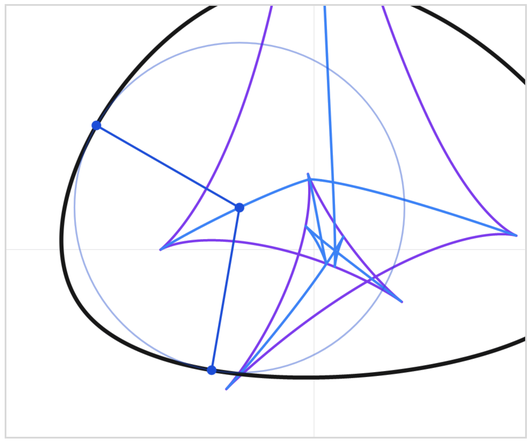
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Symmetry and Focal Sets

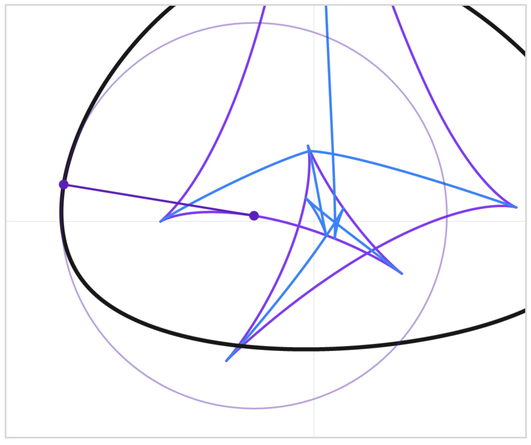
Consider a manifold $\mathcal{M} \subset \mathbb{R}^2$ without boundary.

The symmetry set $\mathrm{sym}(\mathcal{M})$ is the set of centres of all circles tangent to $\mathcal{M}$ in two or more points.



$\mathcal{M}$ $\mathrm{sym}(\mathcal{M})$ two tangency points and the centre they share

The focal set $\mathrm{foc}(\mathcal{M})$ is the set of the centres of the osculating circles — at each point of $\mathcal{M}$, the centre of the circle that has 2nd order contact with the curve.



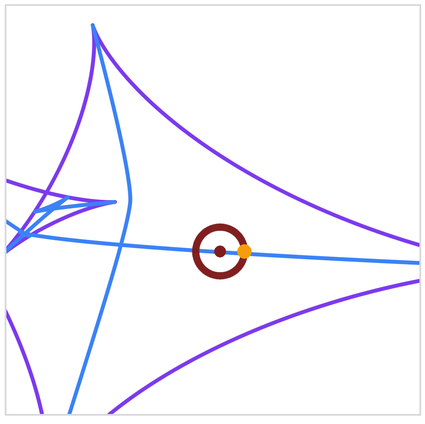
$\mathcal{M}$ $\mathrm{foc}(\mathcal{M})$ contact point and centre of curvature

We refer to the union $\mathrm{foc}(\mathcal{M}) \cup \mathrm{sym}(\mathcal{M})$ as the **bifurcation set**.

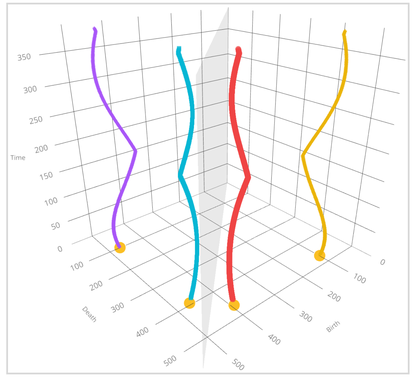
Radial Persistence Transform

Fix a base point x and consider the squared Euclidean distance $d^2_{\mathbb{E}}(\cdot, x)|_{\mathcal{M}}$ function restricted to $\mathcal{M}$. The **RPT** sends x to the **(extended) persistence diagram** of this distance function. By stability, sweeping x along a closed **observation loop** γ, the persistence points move continuously sweeping out curves called **vines**. Stacking these diagrams we get a **closed vineyard**:

$$\mathcal{V}(\mathcal{M}, \gamma) = \bigcup_{t \in S^1} \{t\} \times \mathrm{Dgm}(f_t), \quad f_t = d^2_{\mathbb{E}}(\cdot, \gamma(t))|_{\mathcal{M}}.$$



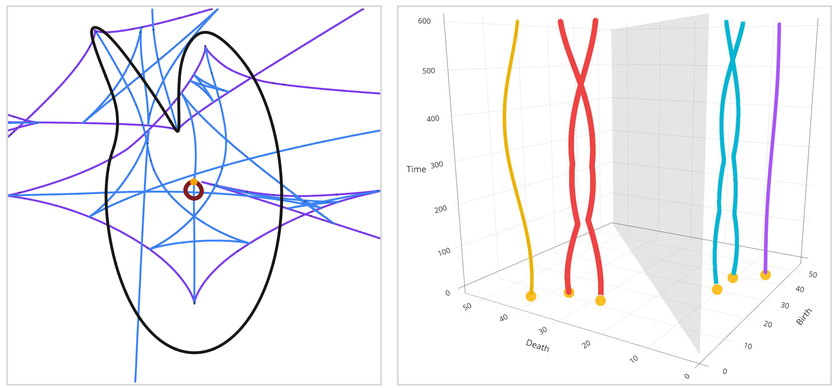
observation loop γ base point x



Bruce–Giblin–Gibson and Porteous: the Morse decomposition of $d^2_{\mathbb{E}}(\cdot, x)|_{\mathcal{M}}$ changes *only* when x crosses the **bifurcation set**.

Monodromy

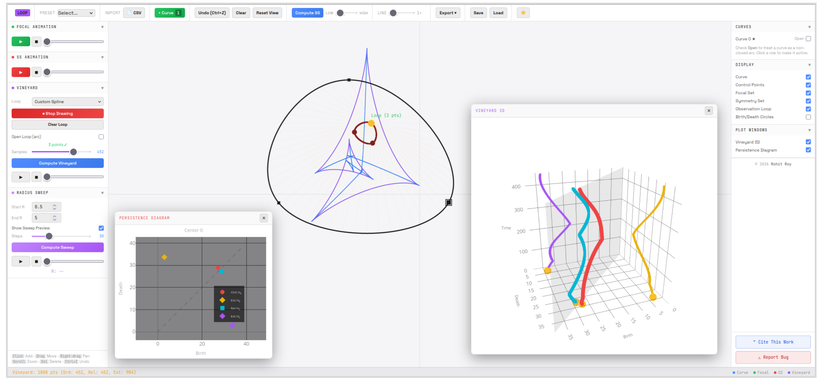
After a full traversal of γ the diagram returns to itself, but its points may come back *permuted*.



Left: a curve with its symmetry and focal sets, and a small observation loop. Right: the vineyard over that loop — following the red vine, we return to a different point, so this loop carries monodromy.

Bouquet2D

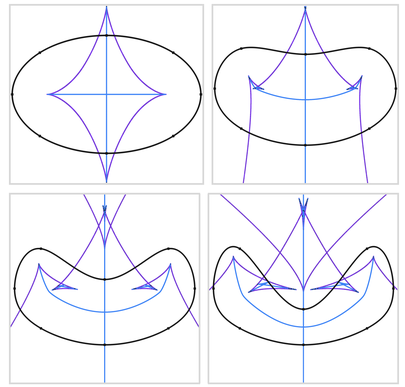
An interactive, browser-based tool:



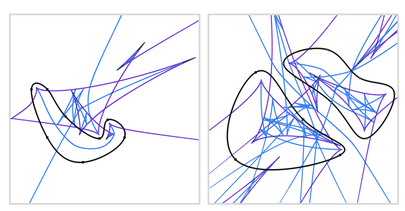
- Draw $\mathcal{M}$ as a spline and drag its control points.
- Symmetry and focal sets are recomputed **in real time** as you deform.
- Place γ as a circle of chosen centre and radius, or draw it freehand.
- Vineyard, 3D vines and diagram update together.
- Extended persistence; several components at once.

Functionality

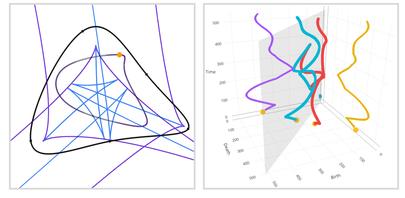
Deform $\mathcal{M}$, and watch the singularities appear.



Arbitrary splines — and several components.



Any observation loop you like, drawn by hand.



Under the Hood

Bitangency. Sample $\mathcal{M}$ and look for sign changes of

$$f_{\pm}(i, j) = (p_i - p_j) \cdot (t_i \pm t_j).$$

Intersect the normals at p_i, p_j for a candidate centre C, kept when $\big|\, \|C - p_i\| - \|C - p_j\| \,\big| < \varepsilon$.

Persistence. Union–find on the lower-star filtration of $d^2_{\mathbb{E}}(\cdot, \gamma(t))|_{\mathcal{M}}$, with the elder rule pairing births to deaths.

[1] Bruce, Giblin, Gibson. *Symmetry sets.* Proc. Roy. Soc. Edinburgh A 104 (1985) 179–204. [2] Porteous. *Geometric differentiation.* CUP, 2001. [3] Cohen-Steiner, Edelsbrunner, Morozov. *Vines and vineyards.* SoCG 2006.
[4] Chambers, Fillmore, Stephenson, Wintraecken. *Braiding vineyards.* SODA 2026. [5] Arya, Giunti, Hickok, Kanari, McGuire, Turner. *Decomposing the PHT of star-shaped objects.* arXiv:2408.14995.

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Feedback very welcome!
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